

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular/Supplementary Winter Examination – 2024****Course: B. Tech Branch: Computer Engineering/Computer Science and Engg(AI & ML) & Allied****Subject Code & Name: Data Structures (BTCOC303)****Semester : III****Max Marks: 60****Date:10/02/2025****Duration: 2 Hrs.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/ CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
	Linked list is not suitable data structure for which one of the following problems? a) Insertion sort b) Binary search c) Radix sort d) Polynomial manipulation		1
	A program P reads in 500 integers in the range [0..100] representing the scores of 500 students. It then prints the frequency of each score above 50. What would be the best way for P to store the frequencies? a) An array of 50 numbers b) An array of 100 numbers c) An array of 500 numbers d) A dynamically allocated array of 550 numbers		1
	You have to sort a list L consisting of a sorted list followed by a few random elements. Which of the following sorting methods would be especially suitable for such a task? a) Bubble sort b) Selection sort c) Quick sort d) Insertion sort		1
4	The number of interchanges required to sort 10, 6, 11, 7,		1

	9 in ascending order using bubble sort is a) 7 b) 5 c) 6 d) 8		
5	The number of element in an array $A[l1:u1, l2:u2]$ is given by a) $(u1-l1-1)(u2-l2-1)$ b) $(u1 \times u2)$ c) $(u1-l1)(u2-l2)$ d) $(u1-l1+1)(u2-l2+1)$		1
6	A situation where one wants to delete data from a data structure that is empty. a) Free storage b) Underflow c) Overflow d) Compaction		1
7	Which of the following is/are application(s) of linked list? a) To implement file systems b) For separate chaining in hash-table c) To implement non-binary trees d) All of the mentioned		1
8	The matrix contains m rows and n columns. The matrix is called sparse matrix if a) Total number of Zero elements $> (m \times n)/2$ b) Total number of Zero elements $= m+n$ c) Total number of Zero elements $= m/n$ d) Total number of Zero elements $= m-n$		1
9	A binary tree with n nodes has null childrens? a) n b) n+1 c) n/2 d) n^2		1
10	The maximum number of edges in an undirected graph with n vertices is a) $n(n-1)/2$ b) 2^n c) n! d) $(n-1)/2$		1
11	How many distinct binary search trees can be constructed with 3 distinct keys? a) 4 b) 5 c) 6 d) 9		1
12	The best data structure to check whether an arithmetic expression has balanced parenthesis is. a) Queue b) Stack c) Tree d) Linked list		1
Q. 2	Solve the following.		12
A)	Why queue is called abstract data type (ADT). Write an		6

	algorithm to implement enqueue and dequeue operations using array as data structure.		
B)	What is binary search tree (bst)? Construct bst for the following data items and show tree after each insert operation. 85, 40, 7, 55, 91, 89, 24		6
Q.3	Solve the following.		12
A)	Convert the infix expression $((A + B) * C - (D - E) ^ (F + G))$ to equivalent Prefix and Postfix notations using stack.		6
B)	Explain how to implement two stacks using one array $A[1..n]$ in such way that neither stack overflows unless the total number of elements in both stacks together is n . The PUSH() and POP() operations should run in $O(1)$ time.		6
Q.4	Solve Any Two of the following.		12
A)	What is hash data structure? Consider the Keys $(k) = \{25, 38, 41, 52, 3, 13, 56\}$ are inserted into an empty hash table using hash function $h(k) = (k \bmod 7)$ and linear probing is used for collision resolution. Give content of the hash table after every key insertion.		6
B)	Consider insertion of characters into empty circular linked list A, B, C, D, E, F. Assume these elements are stored using 10 memory locations. Find the sequence of characters in the list. Suppose F and then C are deleted and G is inserted at the beginning of list. Find the final structure.		6
	What is linked list? Let List be a linked list in memory. Write a separate procedure for each of the following which: i) Finds the how many times a given Item occurs in the List. ii) Add a given value K to each Item in the List.		6

Q.5	Solve Any Two of the following.	12
A)	The inorder and preorder traversals of T yield the following: i) Inorder traversal: E, A, C, K, F, H, D, B, G ii) Preorder traversal: F, A, E, K, C, D, H, G, B Construct the original binary tree T using above two sequences.	6
B)	A scheme for storing binary trees in an array X is as follows. Indexing of an array start with 1. The root is stored at $X[1]$. For a node stored at $X[i]$, the left child is stored in $X[2i]$ and the right child is stored in $X[2i+1]$. Give the array representation of the below given tree. Also, in worst case to store any binary tree of n number of vertices, how many places do you need in array? 	6
	Write and explain bubble sort algorithm. Also, prove that the complexity of bubble sort is $O(n^2)$.	6
Q.6	Solve Any Two of the following.	12
A)	An array $A[1...5][1...5]$ is stored in the memory with each element requiring 4 bytes of storage. If the base address of array is 300, then determine the location of $A[3][4]$, if the array elements are stored (i) Row wise and (ii) Column wise.	6
B)	Enlist properties of the following trees. i) Red black tree iii) AVL tree iv) m-way search tree vi) b+ tree	6
C)	Consider the simple graph given below. Describe G formally in terms of its set V of nodes and its set E of edges. Also find the indegree, outdegree and total degree of each node.	6

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